

A Finite-Block Calculus for the Fabius–Rvachev–Thue–Morse System

New zeta–Lambert tails, q -binomial acceleration, probabilistic deconvolution, and Lambert- W truncation laws

Frontier research report prepared from the complete documentation corpus

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Corpus snapshot: living documents on `main`; frozen historical variants deduplicated through the archive provenance map

Abstract

The documentation tree under `Analysis/FabiusFunction` already contains an unusually broad and internally cross-checked theory of the Thue–Morse sequence, the bounded Fabius distribution function, Rvachev’s compactly supported up-function, dyadic q -binomial arithmetic, infinite sinc products, and the lower-Lambert endpoint phase. This report audits the living manuscripts, the imported primary literature, the active Fourier-decay drafts, and the frozen historical variants, then develops a new finite-block calculus that joins several strands which had previously remained adjacent rather than identical.

The central observation is exact. A single length- 2^m signed Thue–Morse block polynomial, under an oscillatory substitution, is a phase-and-scale normalization of the first m factors of the Fourier sinc product; under a decaying substitution, the same polynomial is a normalization of the first m factors of the negative-Laplace product used in endpoint asymptotics. The omitted tail is not merely small: it is a scaled copy of the full product. This yields (i) a convergent all-orders expansion whose coefficients are zeta values arranged as a Lambert series with base $q = 1/4$; (ii) explicit, sharp-radius remainder bounds; (iii) arbitrary-order finite Thue–Morse approximants to the Fourier transform; (iv) a q -binomial Richardson scheme whose order- p extrapolant converges like $4^{-(p+1)m}$; (v) an exact random-tail decomposition and a weak differential expansion of the up-density in terms of finite uniform splines; (vi) a Lambert- W law for the number of product factors needed on the canonical endpoint saddle scale; and (vii) a Barnes-like tower built from derivatives of the entire shifted Thue–Morse Dirichlet function at its trivial zeros.

All stated theorems are proved here. High-precision computations independently check the finite identities, coefficients, asymptotic orders, and truncation laws. The phrase “new result” means new to the inspected repository corpus and derived in this report; no claim of global publication priority is made without a dedicated literature review beyond the searches recorded here.

Status discipline. The report uses three levels. Statements explicitly attributed to the repository or to the literature are background. Statements labeled “Result derived in this report” are supplied with proofs. Numerical experiments are validation, not substitutes for proof. Several final research directions are labeled as conjectures or formalization obligations.

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1 Corpus audit and research positioning

1.1 What was inspected

The repository’s documentation is not one article but a layered research environment. The living primary exposition separates the bounded cumulative distribution F , the even bump up, and the signed global extension, and develops their construction, probability model, Fourier products, dyadic arithmetic, and corrected endpoint asymptotics. The consolidated frontier volume records non-formalized deductions and gaps; the Thue–Morse Atlas develops a large formula network; the active Fourier-decay directory contains twelve mutually auditing manuscripts; the non-elementarity article and Lean walkthrough provide logical and formal context; and the papers directory contains source TeX for relevant primary literature. The archive is explicitly a provenance-preserving collection of frozen versions, with byte-identical or otherwise redundant entries pruned from the historical catalogue [1, 2, 3, 4, 5].

The audit protocol was therefore:

1. read the current living manuscripts as the authoritative mathematical state;
2. compare the active Fourier-decay drafts because they contain genuinely different hypotheses, corrections, and audit conclusions;
3. inspect each imported paper represented by TeX source;
4. use the archive’s commit-level provenance map to deduplicate superseded historical copies, while retaining every uniquely described line of development in the research map;
5. search the corpus for the candidate identities and coefficient patterns before calling them new to the corpus.

A complete source ledger appears in [section C](#).

1.2 Established pillars

The following table compresses the established state which this report uses but does not re-prove in full.

Pillar	Established content	Remaining junction exploited here
Thue–Morse	$\varepsilon_n = (-1)^{s_2(n)}$, block products, Prouhet cancellation, Mahler equations, Walsh and lacunary products, shifted entire Dirichlet theory	Turn one finite block into a certified approximation engine for both Fourier and Laplace products
Fabius/up probability	up as a compactly supported probability density and F as its rescaled CDF; finite atomic/uniform convolution models	Identify the omitted convolution tail as an exact scaled copy and expand it to all orders
Fourier analysis	$\widehat{\text{up}}$ as a dyadic sinc product; shell factorization; pointwise, metric, averaged, and extremal decay regimes	Analyze truncation in the small tail variable, which is complementary to large-frequency shell dynamics
Dyadic arithmetic	exact values at dyadic rationals, q -Pochhammer and Gaussian-binomial formulae at base $1/2$, Thue–Morse coefficient extraction	Produce a new Gaussian-binomial role at base $q = 1/4$: geometric-grid extrapolation weights
Endpoint asymptotics	negative-Laplace product, periodic lower-Lambert phase, all-orders saddle expansion, $\lambda = -W_{-1}(-x \log 2)/\log 2$	Quantify the finite-product depth on the canonical Lambert scale and express it through one finite Thue–Morse block
Formalization	extensive Lean construction and exact proof correspondence; explicit quarantine of unformalized claims	Isolate short algebraic and analytic lemmas suitable for promotion

1.3 The missing commutative bridge

The corpus contains all four vertices of the following conceptual square:

$$\begin{array}{ccc} \text{finite Thue–Morse blocks} & \longrightarrow & \text{finite dyadic products} \\ \downarrow & & \downarrow \\ \text{exact dyadic arithmetic} & \longrightarrow & \text{Fourier/Laplace asymptotics.} \end{array}$$

What was not isolated as one theorem package is that the top horizontal arrow has two exact specializations, oscillatory and decaying, and that the right vertical arrow has a self-similar tail with an analytic expansion in the geometric parameter 4^{-m} . Once this is written down, zeta values, Lambert series, Gaussian binomials, random-tail moments, and the Lambert- W endpoint phase become parts of one finite-block calculus rather than separate phenomena.

1.4 Priority and uncertainty

Keyword searches of the inspected TeX corpus did not locate the all-orders 4^{-m} tail formula, the sharp-radius remainder bound, or the q -binomial extrapolation scheme in the form proved below. Searches of arXiv and publisher records located the classical Fabius, Thue–Morse, lacunary-product, automatic-sequence, arithmetic, Dirichlet-series, and Lambert- W sources used for context [6, 7, 8, 9, 10, 11], but did not constitute an exhaustive priority search. Accordingly, every novelty statement in this report is corpus-relative.

2 Objects, normalizations, and exact background

2.1 The signed block polynomial

Let

$$\varepsilon_n = (-1)^{s_2(n)}, \quad P_m(z) = \sum_{n=0}^{2^m-1} \varepsilon_n z^n = \prod_{j=0}^{m-1} (1 - z^{2^j}), \quad (2.1)$$

where $s_2(n)$ is the binary digit sum. The product identity is the finite form of the Thue–Morse generating-product law. Its zero of order m at $z = 1$ is exactly the Prouhet cancellation of moments of degrees below m .

2.2 Fourier and Laplace products

Use $\text{sinc } z = \sin z/z$ with $\text{sinc } 0 = 1$, and define

$$Q_m(t) = \prod_{k=1}^m \text{sinc}\left(\frac{t}{2^k}\right), \quad Q_\infty(t) = \prod_{k=1}^{\infty} \text{sinc}\left(\frac{t}{2^k}\right), \quad (2.2)$$

$$G_m(-s) = \prod_{k=1}^m \frac{1 - e^{-s/2^k}}{s/2^k}, \quad G(-s) = \prod_{k=1}^{\infty} \frac{1 - e^{-s/2^k}}{s/2^k}. \quad (2.3)$$

The products converge locally uniformly and define entire functions after removable values at zero are supplied.

With the Fourier convention

$$\widehat{h}(\xi) = \int_{\mathbb{R}} h(x) e^{-2\pi i x \xi} dx,$$

the normalized Rvachev density satisfies

$$\widehat{\text{up}}(\xi) = Q_\infty(2\pi\xi) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi\xi}{2^j}\right). \quad (2.4)$$

The bounded Fabius distribution is related by

$$F'(x) = 2 \text{up}(2x - 1), \quad F(x) = \int_{-1}^{2x-1} \text{up}(y) dy \quad (0 \leq x \leq 1). \quad (2.5)$$

These conventions agree with the repository's primary exposition [1, 6].

2.3 The lower-Lambert phase

Write $L = \log 2$. For small $x > 0$, the canonical endpoint phase variable is

$$\lambda(x) = -\frac{W_{-1}(-Lx)}{L}, \quad x = \lambda(x)2^{-\lambda(x)}. \quad (2.6)$$

The associated leading saddle scale is

$$s_0(x) = 2^{\lambda(x)} = \frac{\lambda(x)}{x}. \quad (2.7)$$

The repository's endpoint theory adds periodic corrections in $\log_2 s$ and an all-orders saddle expansion. Our use of (2.7) below is computational: it measures how many factors of the exact product are needed near that scale. It is not a replacement for the periodically corrected saddle analysis.

3 One Thue–Morse block, two exact product bridges

3.1 Oscillatory specialization: the Fourier bridge

Theorem 3.1 (Exact finite Thue–Morse–sinc identity). *For every integer $m \geq 1$ and every $t \in \mathbb{C}$,*

$$P_m\left(e^{it/2^{m-1}}\right) = (-i)^m t^m 2^{-\binom{m}{2}} e^{it(1-2^{-m})} Q_m(t). \quad (3.1)$$

Equivalently, with the removable value at $t = 0$ understood,

$$Q_m(t) = i^m 2^{\binom{m}{2}} t^{-m} e^{-it(1-2^{-m})} \sum_{n=0}^{2^m-1} \varepsilon_n e^{int/2^{m-1}}. \quad (3.2)$$

Result derived in this report.

Proof. Insert $z = e^{it/2^{m-1}}$ in the product form of P_m and reverse the index:

$$P_m(z) = \prod_{k=0}^{m-1} \left(1 - e^{it/2^k}\right).$$

For each factor use

$$1 - e^{ix} = (-i)xe^{ix/2} \text{sinc}(x/2).$$

The product of the linear terms is

$$(-i)^m t^m 2^{-\sum_{k=0}^{m-1} k} = (-i)^m t^m 2^{-\binom{m}{2}}.$$

The product of the phases is

$$\exp\left(\frac{it}{2} \sum_{k=0}^{m-1} 2^{-k}\right) = e^{it(1-2^{-m})},$$

and the sinc factors are $\text{sinc}(t/2), \dots, \text{sinc}(t/2^m)$. This gives (3.1); the sum form follows from (2.1). $\square$

Remark 3.2 (Prouhet cancellation becomes the normalization). The factor t^{-m} in (3.2) is harmless precisely because P_m has a zero of order m at 1. Thus the same cancellation that annihilates the first m power moments also regularizes the Fourier approximation at the origin.

3.2 Decaying specialization: the endpoint Laplace bridge

Theorem 3.3 (Exact finite Thue–Morse–Laplace identity). *For every integer $m \geq 1$ and every $s \in \mathbb{C}$,*

$$G_m(-s) = \frac{2^{m(m+1)/2}}{s^m} P_m\left(e^{-s/2^m}\right) = \frac{2^{m(m+1)/2}}{s^m} \sum_{n=0}^{2^m-1} \varepsilon_n e^{-ns/2^m}. \quad (3.3)$$

Again, the apparent singularity at $s = 0$ is removable. **Result derived in this report.**

Proof. The substitution $z = e^{-s/2^m}$ gives

$$P_m(z) = \prod_{j=0}^{m-1} \left(1 - e^{-s2^j/2^m}\right) = \prod_{k=1}^m \left(1 - e^{-s/2^k}\right).$$

The denominator in $G_m(-s)$ is

$$\prod_{k=1}^m \frac{s}{2^k} = s^m 2^{-m(m+1)/2},$$

which proves the identity. $\square$

The exact finite-block principle. The oscillatory sample $z = e^{it/2^{m-1}}$ converts P_m into the truncated Fourier product $Q_m(t)$. The decaying sample $z = e^{-s/2^m}$ converts the same P_m into the truncated negative-Laplace product $G_m(-s)$. No asymptotic argument is involved.

3.3 The omitted tail is a scaled copy of the whole

Proposition 3.4 (Exact tail self-similarity). *For every $m \geq 0$,*

$$Q_\infty(t) = Q_m(t) Q_\infty(t/2^m), \quad (3.4)$$

$$G(-s) = G_m(-s) G(-s/2^m). \quad (3.5)$$

Result derived in this report.

Proof. After removing the first m factors, replace k by $m + \ell$ in the remaining infinite product. The residual factors are exactly those of the original product evaluated at the scaled argument. $\square$

Corollary 3.5 (Exact finite Thue–Morse factorization of the full products). *For $m \geq 1$,*

$$Q_\infty(t) = i^m 2^{\binom{m}{2}} t^{-m} e^{-it(1-2^{-m})} P_m\left(e^{it/2^{m-1}}\right) Q_\infty(t/2^m), \quad (3.6)$$

$$G(-s) = \frac{2^{m(m+1)/2}}{s^m} P_m\left(e^{-s/2^m}\right) G(-s/2^m). \quad (3.7)$$

Thus every truncation error is represented by the original entire function in a small argument.

4 The zeta–Lambert tail calculus

4.1 A local logarithm of the sinc function

Lemma 4.1 (Euler–zeta expansion). *For $|z| < \pi$,*

$$\log \operatorname{sinc} z = - \sum_{r=1}^{\infty} \frac{\zeta(2r)}{r\pi^{2r}} z^{2r}. \quad (4.1)$$

The logarithm is the branch vanishing at $z = 0$.

Proof. Euler’s product gives

$$\frac{\sin z}{z} = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{\pi^2 n^2}\right).$$

For $|z| < \pi$, expand $\log(1 - w) = -\sum_{r \geq 1} w^r / r$. Absolute local uniform convergence permits interchanging the sums, and the inner sum over n is $\zeta(2r)$. $\square$

4.2 All orders, exact radius, explicit remainder

Theorem 4.2 (All-orders sinc-tail expansion). *Let $m \geq 0$. For*

$$|t| < \pi 2^{m+1}, \quad (4.2)$$

the quotient $Q_{\infty}(t)/Q_m(t)$ is nonzero and

$$\log \frac{Q_{\infty}(t)}{Q_m(t)} = - \sum_{r=1}^{\infty} \frac{\zeta(2r)}{r\pi^{2r}(4^r - 1)} t^{2r} 4^{-rm}. \quad (4.3)$$

*The disk in (4.2) is maximal for this Taylor logarithm: its boundary contains the nearest zeros $t = \pm \pi 2^{m+1}$ of the tail. **Result derived in this report.***

Proof. By [theorem 3.4](#), the quotient is $Q_{\infty}(t/2^m)$. Apply [theorem 4.1](#) to every factor and sum:

$$\begin{aligned} \log Q_{\infty}(t/2^m) &= - \sum_{k=1}^{\infty} \sum_{r=1}^{\infty} \frac{\zeta(2r)}{r\pi^{2r}} \left(\frac{t}{2^{m+k}}\right)^{2r} \\ &= - \sum_{r=1}^{\infty} \frac{\zeta(2r)t^{2r}}{r\pi^{2r}} \frac{4^{-r(m+1)}}{1 - 4^{-r}} \\ &= - \sum_{r=1}^{\infty} \frac{\zeta(2r)}{r\pi^{2r}(4^r - 1)} t^{2r} 4^{-rm}. \end{aligned}$$

The closest zero of $Q_{\infty}(w)$ to $w = 0$ is $w = \pm 2\pi$, arising from the factor $\operatorname{sinc}(w/2)$. Scaling by $w = t/2^m$ proves maximality. $\square$

The first terms are

$$\frac{Q_{\infty}(t)}{Q_m(t)} = \exp \left(-\frac{t^2}{18} 4^{-m} - \frac{t^4}{2700} 16^{-m} - \frac{t^6}{178605} 64^{-m} - \frac{t^8}{9639000} 256^{-m} - \dots \right). \quad (4.4)$$

Theorem 4.3 (Certified tail remainder). *Let $N \geq 0$ and put*

$$x_m(t) = \frac{|t|}{\pi 2^{m+1}} < 1. \quad (4.5)$$

If $R_{N,m}(t)$ denotes the part of the positive series in (4.3) with $r \geq N + 1$, then

$$|R_{N,m}(t)| \leq \frac{4\zeta(2)}{3} \frac{x_m(t)^{2N+2}}{(N+1)(1-x_m(t)^2)}. \quad (4.6)$$

For real t in the stated disk, $R_{N,m}(t) \geq 0$. **Result derived in this report.**

Proof. Rewrite the absolute value of the r -th term as

$$\frac{\zeta(2r)}{r} \frac{4^r}{4^r - 1} x_m(t)^{2r}.$$

Use $\zeta(2r) \leq \zeta(2)$, $4^r/(4^r - 1) \leq 4/3$, and $1/r \leq 1/(N+1)$ for $r \geq N+1$, then sum the geometric series. $\square$

4.3 Arbitrary-order finite Thue–Morse approximants

Definition 4.4 (Zeta-corrected block approximant). For $N \geq 0$ define

$$Q_m^{[N]}(t) = Q_m(t) \exp \left[- \sum_{r=1}^N \frac{\zeta(2r)t^{2r}}{r\pi^{2r}(4^r - 1)} 4^{-rm} \right]. \quad (4.7)$$

For $N = 0$, this is Q_m .

Corollary 4.5 (Arbitrary-order convergence). For every compact set $K \subset \mathbb{C}$ and every fixed $N \geq 0$,

$$Q_m^{[N]}(t) = Q_\infty(t) \left(1 + \mathcal{O}_K(4^{-(N+1)m}) \right) \quad (t \in K, m \rightarrow \infty). \quad (4.8)$$

Moreover, after inserting (3.2), the left side is a normalized length- 2^m Thue–Morse exponential sum multiplied by only N explicit zeta corrections. **Result derived in this report.**

Proof. For all sufficiently large m , K lies in the disk (4.2). The exact quotient $Q_\infty/Q_m^{[N]}$ is $\exp(-R_{N,m})$. The bound (4.6), uniformly on K , is $\mathcal{O}_K(4^{-(N+1)m})$; exponentiation preserves that order. $\square$

Corollary 4.6 (Fourier approximation to the up-function). For fixed $\xi \in \mathbb{C}$,

$$\begin{aligned} \widehat{\text{up}}(\xi) &= i^m 2^{\binom{m}{2}} (2\pi\xi)^{-m} e^{-2\pi i \xi(1-2^{-m})} \sum_{n=0}^{2^m-1} \varepsilon_n e^{2\pi i n \xi / 2^{m-1}} \\ &\times \exp \left[- \sum_{r=1}^N \frac{\zeta(2r)(2\pi\xi)^{2r}}{r\pi^{2r}(4^r - 1)} 4^{-rm} \right] \left(1 + \mathcal{O}_{\xi,N}(4^{-(N+1)m}) \right). \end{aligned} \quad (4.9)$$

Thus the infinite Fourier product admits an arbitrary-order approximation from one finite digital block.

4.4 Why the coefficient array is a Lambert series

Put $q = 1/4$. Then (4.3) becomes

$$\log \frac{Q_\infty(t)}{Q_m(t)} = - \sum_{r=1}^{\infty} \frac{\zeta(2r)}{r\pi^{2r}} t^{2r} \frac{q^{r(m+1)}}{1 - q^r}. \quad (4.10)$$

The kernel $q^r/(1 - q^r)$ is the elementary Lambert-series kernel. Expanding it as $\sum_{\ell \geq 1} q^{r\ell}$ reconstructs the omitted factors. This is a direct analytic connection between the sinc product and q -series: the base is $q = 1/4$ because the logarithm is even and the geometric frequency scale is $1/2$.

Theorem 4.7 (General geometric base). *Let $b > 1$ and*

$$Q_{m,b}(t) = \prod_{k=1}^m \text{sinc}(t/b^k), \quad Q_{\infty,b}(t) = \prod_{k=1}^{\infty} \text{sinc}(t/b^k), \quad q = b^{-2}.$$

For $|t| < \pi b^{m+1}$,

$$\log \frac{Q_{\infty,b}(t)}{Q_{m,b}(t)} = - \sum_{r=1}^{\infty} \frac{\zeta(2r)}{r\pi^{2r}} t^{2r} \frac{q^{r(m+1)}}{1 - q^r}. \quad (4.11)$$

Result derived in this report.

Proof. Repeat the proof of [theorem 4.2](#), replacing the geometric sum in 4^{-r} by the one in b^{-2r} . $\square$

5 An all-orders finite-block formula for the endpoint Laplace product

5.1 Centering removes the linear tail

Lemma 5.1 (Fourier–Laplace rotation). *For every $s \in \mathbb{C}$,*

$$G(-s) = e^{-s/2} Q_{\infty}(is/2). \quad (5.1)$$

More generally,

$$G_m(-s) = e^{-\frac{s}{2}(1-2^{-m})} Q_m(is/2). \quad (5.2)$$

Proof. For every v ,

$$\frac{1 - e^{-v}}{v} = e^{-v/2} \frac{\sinh(v/2)}{v/2} = e^{-v/2} \text{sinc}(iv/2).$$

Multiply with $v = s/2^k$. The linear exponents sum to $-s/2$ in the infinite product and to $-s(1 - 2^{-m})/2$ in the finite product. $\square$

Define the centered products

$$C(-s) = e^{s/2} G(-s), \quad C_m(-s) = e^{\frac{s}{2}(1-2^{-m})} G_m(-s). \quad (5.3)$$

Then $C(-s) = Q_{\infty}(is/2)$ and

$$C(-s) = C_m(-s) C(-s/2^m). \quad (5.4)$$

The centering is not cosmetic: the raw tail has a first-order error, while the centered tail has a second-order error.

5.2 Bernoulli–zeta expansion of the negative tail

Theorem 5.2 (All-orders negative-Laplace tail). *For $|u| < 4\pi$,*

$$\log G(-u) = -\frac{u}{2} + \sum_{r=1}^{\infty} \frac{B_{2r}}{2r(2r)!(4^r - 1)} u^{2r}. \quad (5.5)$$

Equivalently,

$$\log C(-u) = \frac{u^2}{72} - \frac{u^4}{43200} + \frac{u^6}{11430720} - \frac{u^8}{2467584000} + \cdots \quad (5.6)$$

The radius 4π is exact. **Result derived in this report.**

Proof. Use [theorem 5.1](#) and [theorem 4.2](#) with $m = 0$ and $t = iu/2$:

$$\log G(-u) = -\frac{u}{2} - \sum_{r \geq 1} \frac{\zeta(2r)}{r\pi^{2r}(4^r - 1)} (-1)^r \frac{u^{2r}}{4^r}.$$

The standard evaluation

$$\zeta(2r) = (-1)^{r+1} \frac{B_{2r}(2\pi)^{2r}}{2(2r)!}$$

reduces the coefficient to the one displayed. The closest zeros of $G(-u)$ are $u = \pm 4\pi i$, coming from $1 - e^{-u/2}$, so the logarithmic radius is exact. $\square$

Corollary 5.3 (Certified remainder for the centered Laplace tail). *Let $y = |u|/(4\pi) < 1$. If the series in (5.5) is stopped after u^{2N} , while the exact linear term $-u/2$ is retained, then the remainder $\mathcal{R}_N(u)$ obeys*

$$|\mathcal{R}_N(u)| \leq \frac{4\zeta(2)}{3} \frac{y^{2N+2}}{(N+1)(1-y^2)}. \quad (5.7)$$

Proof. This is [theorem 4.3](#) at $m = 0$, $t = iu/2$. $\square$

5.3 The endpoint product from one finite digital block

Theorem 5.4 (Finite Thue–Morse endpoint formula with all corrections). *Let $m \geq 1$, $s \in \mathbb{C}$, and $u = s/2^m$. If $|u| < 4\pi$, then*

$$\boxed{G(-s) = \frac{2^{m(m+1)/2}}{s^m} P_m(e^{-u}) \times \exp \left[-\frac{u}{2} + \sum_{r=1}^{\infty} \frac{B_{2r}}{2^r (2r)! (4^r - 1)} u^{2r} \right]}. \quad (5.8)}$$

Stopping the correction series after $r = N$ incurs the explicit logarithmic error (5.7). In sum form, $P_m(e^{-u}) = \sum_{n < 2^m} \varepsilon_n e^{-nu}$. **Result derived in this report.**

Proof. Combine the exact factorization (3.7) with [theorem 5.2](#) applied to the tail $G(-u)$. $\square$

Remark 5.5 (Arithmetic-to-asymptotic bridge). The exact dyadic arithmetic literature typically uses finite coefficient extraction and q -binomial identities, whereas endpoint asymptotics use the infinite product $G(-s)$ and a saddle. Formula (5.8) gives an intermediate object: a single finite Thue–Morse exponential sum, multiplied by a convergent universal correction whose coefficients are Bernoulli or zeta values. It is exact within its natural logarithmic disk, not merely asymptotic in m .

6 q -binomial Richardson acceleration on the spectral scale

6.1 An analytic error germ in q^m

Put $q = 1/4$. From (3.4),

$$Q_m(t) = Q_\infty(t) H_t(q^m), \quad H_t(r) = \frac{1}{Q_\infty(t\sqrt{r})}. \quad (6.1)$$

Although a square root appears in the notation, Q_∞ is even, so H_t is a genuine meromorphic function of r . Its Taylor expansion at 0 is

$$H_t(r) = \exp \left(\sum_{\nu \geq 1} \frac{\zeta(2\nu) t^{2\nu}}{\nu \pi^{2\nu} (4^\nu - 1)} r^\nu \right) = \sum_{n \geq 0} h_n(t) r^n. \quad (6.2)$$

For $t \neq 0$, the nearest pole occurs at $r = 4\pi^2/t^2$. The first coefficients are

$$h_0 = 1, \quad h_1 = \frac{t^2}{18}, \quad h_2 = \frac{31t^4}{16200}, \quad h_3 = \frac{2347t^6}{42865200}. \quad (6.3)$$

6.2 Geometric-grid weights are Gaussian binomials

For $(q; q)_n = \prod_{k=1}^n (1 - q^k)$, define

$$w_{p,j} = \frac{(-1)^{p-j} q^{\binom{p-j+1}{2}}}{(q; q)_j (q; q)_{p-j}} = \frac{(-1)^{p-j} q^{\binom{p-j+1}{2}}}{(q; q)_p} \begin{bmatrix} p \\ j \end{bmatrix}_q, \quad 0 \leq j \leq p. \quad (6.4)$$

Lemma 6.1 (Lagrange characterization). *The numbers in (6.4) are the Lagrange weights for evaluating at $r = 0$ from the geometric nodes $1, q, \dots, q^p$:*

$$w_{p,j} = \prod_{\substack{0 \leq \ell \leq p \\ \ell \neq j}} \frac{-q^\ell}{q^j - q^\ell}. \quad (6.5)$$

Consequently,

$$\sum_{j=0}^p w_{p,j} = 1, \quad \sum_{j=0}^p w_{p,j} q^{jn} = 0 \quad (1 \leq n \leq p), \quad (6.6)$$

and

$$\sum_{j=0}^p w_{p,j} q^{j(p+1)} = (-1)^p q^{p(p+1)/2}. \quad (6.7)$$

Proof. Split the denominator in (6.5) into $\ell < j$ and $\ell > j$, factor the powers of q , and collect the factors $1 - q^k$. This gives (6.4). The moment identities state that Lagrange interpolation reproduces the monomials $1, r, \dots, r^p$ at $r = 0$. For r^{p+1} , the interpolation error is the nodal polynomial $\prod_{\ell=0}^p (r - q^\ell)$; evaluation at 0 gives $(-1)^{p+1} q^{p(p+1)/2}$ for $f = P$, hence (6.7) for $P = f$. $\square$

Theorem 6.2 (q -binomial spectral Richardson extrapolation). *Define*

$$\mathcal{R}_{m,p}(t) = \sum_{j=0}^p w_{p,j} Q_{m+j}(t), \quad q = \frac{1}{4}. \quad (6.8)$$

For every compact $K \subset \mathbb{C}$ and fixed p ,

$$\boxed{\mathcal{R}_{m,p}(t) = Q_\infty(t) + \mathcal{O}_K(q^{(p+1)m}) \quad (t \in K).} \quad (6.9)$$

More precisely,

$$\mathcal{R}_{m,p}(t) - Q_\infty(t) = Q_\infty(t) (-1)^p q^{p(p+1)/2} h_{p+1}(t) q^{(p+1)m} + \mathcal{O}_K(q^{(p+2)m}). \quad (6.10)$$

Every Q_{m+j} may be replaced by its exact finite Thue–Morse representation (3.2); hence (6.8) is a finite digital extrapolation formula with Gaussian-binomial coefficients. **Result derived in this report.**

Proof. Insert (6.2) into (6.8):

$$\begin{aligned} \frac{\mathcal{R}_{m,p}(t)}{Q_\infty(t)} &= \sum_{j=0}^p w_{p,j} \sum_{n \geq 0} h_n(t) q^{n(m+j)} \\ &= \sum_{n \geq 0} h_n(t) q^{nm} \left(\sum_{j=0}^p w_{p,j} q^{jn} \right). \end{aligned}$$

The inner sums are 1 at $n = 0$, vanish for $1 \leq n \leq p$, and equal (6.7) at $n = p + 1$. Local uniform convergence of the analytic H_t series gives the uniform remainder on compact t -sets. $\square$

Corollary 6.3 (A derivative bound version). *Suppose $t \in \mathbb{R}$ and the interval $[0, q^m]$ contains no pole of H_t . Then for some $\theta \in (0, q^m)$,*

$$\frac{\mathcal{R}_{m,p}(t)}{Q_\infty(t)} - 1 = \frac{(-1)^p q^{(p+1)m+p(p+1)/2}}{(p+1)!} H_t^{(p+1)}(\theta). \quad (6.11)$$

In particular, a supremum bound on $H_t^{(p+1)}$ gives a fully certified extrapolation error.

Proof. Apply the real Lagrange interpolation remainder to H_t at the nodes $q^m, q^{m+1}, \dots, q^{m+p}$ and evaluate the interpolating polynomial at zero. $\square$

The first four weight rows are

$$\begin{aligned} p=1 : & \quad \left(-\frac{1}{3}, \frac{4}{3} \right), \\ p=2 : & \quad \left(\frac{1}{45}, -\frac{4}{9}, \frac{64}{45} \right), \\ p=3 : & \quad \left(-\frac{1}{2835}, \frac{4}{135}, -\frac{64}{135}, \frac{4096}{2835} \right), \\ p=4 : & \quad \left(\frac{1}{722925}, -\frac{4}{8505}, \frac{64}{2025}, -\frac{4096}{8505}, \frac{1048576}{722925} \right). \end{aligned} \quad (6.12)$$

Remark 6.4 (Two complementary accelerations). The zeta correction (4.7) uses one block level and analytic coefficients. The q -Richardson formula (6.8) uses $p+1$ consecutive block levels but no zeta values. Both exploit the same analytic germ in 4^{-m} , so their observed orders agree: correction through N and extrapolation degree $p = N$ both leave order $4^{-(N+1)m}$.

7 Exact random tails and a weak differential expansion

7.1 The random-series model

Let $V_1, V_2, \dots$ be independent random variables uniformly distributed on $[-1, 1]$, and put

$$Y_m = \sum_{k=1}^m 2^{-k} V_k, \quad Y_\infty = \sum_{k=1}^{\infty} 2^{-k} V_k. \quad (7.1)$$

Then Y_∞ has density up , while Y_m has the compactly supported centered uniform spline density u_m . Their characteristic functions are Q_∞ and Q_m .

Theorem 7.1 (Exact random-tail decomposition). *There is an independent copy Y'_∞ of Y_∞ such that*

$$Y_\infty \stackrel{d}{=} Y_m + 2^{-m} Y'_\infty. \quad (7.2)$$

Consequently, in density form,

$$\text{up} = u_m * (2^m \text{up}(2^m \cdot)). \quad (7.3)$$

Result derived in this report.

Proof. Split the infinite sum after m and reindex:

$$\sum_{k=m+1}^{\infty} 2^{-k} V_k = 2^{-m} \sum_{\ell=1}^{\infty} 2^{-\ell} V_{m+\ell}.$$

The reindexed series is independent of Y_m and has the law of Y_∞ . Convolution of the corresponding densities gives (7.3). $\square$

7.2 Cumulants are the same zeta–Lambert coefficients

Let κ_j and μ_j denote the cumulants and moments of Y_∞ .

Theorem 7.2 (Closed cumulant formula). *All odd cumulants vanish, and for $r \geq 1$,*

$$\kappa_{2r} = (-1)^{r+1} \frac{(2r)! \zeta(2r)}{r \pi^{2r} (4^r - 1)} = \frac{2^{2r-1} B_{2r}}{r(4^r - 1)}. \quad (7.4)$$

In particular,

$$\kappa_2 = \frac{1}{9}, \quad \kappa_4 = -\frac{2}{225}, \quad \kappa_6 = \frac{16}{3969}, \quad \kappa_8 = -\frac{16}{3825}. \quad (7.5)$$

Result derived in this report.

Proof. The cumulant expansion of the characteristic function is

$$\log \mathbb{E} e^{itY_\infty} = \sum_{r \geq 1} \kappa_{2r} \frac{(it)^{2r}}{(2r)!}.$$

Set $m = 0$ in (4.3) and identify coefficients. The Bernoulli form follows from the classical evaluation of $\zeta(2r)$. $\square$

The first even moments, obtained from the complete Bell polynomials in the cumulants, are

$$\mu_2 = \frac{1}{9}, \quad \mu_4 = \frac{19}{675}, \quad \mu_6 = \frac{583}{59535}, \quad \mu_8 = \frac{132809}{32531625}. \quad (7.6)$$

7.3 A certified weak expansion

Theorem 7.3 (Weak differential expansion of the up-law). *Let $N \geq 0$ and $\varphi \in C^{2N+2}([-1, 1])$. Then*

$$\mathbb{E}\varphi(Y_\infty) = \sum_{j=0}^N \frac{\mu_{2j}}{(2j)!} 4^{-jm} \mathbb{E}\varphi^{(2j)}(Y_m) + \mathcal{E}_{N,m}(\varphi), \quad (7.7)$$

where $\mu_0 = 1$ and

$$|\mathcal{E}_{N,m}(\varphi)| \leq \frac{4^{-(N+1)m}}{(2N+2)!} \|\varphi^{(2N+2)}\|_{L^\infty([-1,1])}. \quad (7.8)$$

Result derived in this report.

Proof. Condition on Y_m in (7.2) and Taylor-expand $\varphi(Y_m + 2^{-m}Y'_\infty)$ through degree $2N+1$. Symmetry kills every odd moment. Since $|Y'_\infty| \leq 1$, the Taylor remainder is bounded by $2^{-(2N+2)m} \|\varphi^{(2N+2)}\|_\infty / (2N+2)!$. $\square$

The first terms are therefore

$$\begin{aligned} \mathbb{E}\varphi(Y_\infty) &= \mathbb{E}\varphi(Y_m) + \frac{4^{-m}}{18} \mathbb{E}\varphi''(Y_m) + \frac{19 \cdot 16^{-m}}{16200} \mathbb{E}\varphi^{(4)}(Y_m) \\ &\quad + \frac{583 \cdot 64^{-m}}{42865200} \mathbb{E}\varphi^{(6)}(Y_m) + \mathcal{O}_\varphi(256^{-m}). \end{aligned} \quad (7.9)$$

Corollary 7.4 (Distributional spline expansion). *In the sense of distributions, for every fixed N ,*

$$\text{up} = \sum_{j=0}^N \frac{\mu_{2j}}{(2j)!} 4^{-jm} D^{2j} u_m + \mathcal{O}_{\mathcal{D}'}(4^{-(N+1)m}). \quad (7.10)$$

In particular,

$$\text{up} = u_m + \frac{4^{-m}}{18} D^2 u_m + \frac{19 \cdot 16^{-m}}{16200} D^4 u_m + \dots \quad (7.11)$$

Proof. Pair the asserted distribution with a test function and use $\langle D^{2j}u_m, \varphi \rangle = \mathbb{E}\varphi^{(2j)}(Y_m)$. $\square$

Remark 7.5 (Cumulant operator). Formally, and exactly at the level of the local Fourier logarithm,

$$\text{up} = \exp\left(\sum_{r \geq 1} \frac{\kappa_{2r}}{(2r)!} 4^{-rm} D^{2r}\right) u_m. \quad (7.12)$$

Expanding the exponential changes cumulants into moments and recovers (7.10). This is a deconvolution calculus for the finite uniform splines already present in the repository's repeated-integration and probabilistic constructions.

8 Lambert- W complexity of endpoint product truncation

8.1 The canonical scale and the tail variable

For $0 < x < 1/(eL)$, let $\lambda(x)$ and $s_0(x)$ be given by (2.6)–(2.7). At an integer truncation level m , the entire omitted product is governed by the single tail variable

$$u_{x,m} = \frac{s_0(x)}{2^m} = 2^{\lambda(x)-m}. \quad (8.1)$$

If $m = \lceil \lambda(x) + h \rceil$, then

$$2^{-h-1} < u_{x,m} \leq 2^{-h}. \quad (8.2)$$

Thus each extra factor beyond the Lambert phase halves the raw tail variable.

Theorem 8.1 (Raw and centered depth laws). *Let*

$$\delta_{\text{raw}}(u) = |G(-u) - 1|, \quad (8.3)$$

$$\delta_{\text{ctr}}(u) = |e^{u/2}G(-u) - 1| = |C(-u) - 1|. \quad (8.4)$$

As $u \rightarrow 0^+$,

$$\delta_{\text{raw}}(u) = \frac{u}{2} + \mathcal{O}(u^2), \quad \delta_{\text{ctr}}(u) = \frac{u^2}{72} + \mathcal{O}(u^4). \quad (8.5)$$

Consequently, for $m = \lceil \lambda(x) + h \rceil$ and $h \rightarrow \infty$,

$$\delta_{\text{raw}}(u_{x,m}) = \Theta(2^{-h}), \quad \delta_{\text{ctr}}(u_{x,m}) = \Theta(4^{-h}), \quad (8.6)$$

with absolute two-sided constants once h is sufficiently large. If $m_{\text{raw}}(x, \eta)$ and $m_{\text{ctr}}(x, \eta)$ denote the least depths which make the respective errors at most η , then as $\eta \downarrow 0$,

$$m_{\text{raw}}(x, \eta) = -\frac{W_{-1}(-Lx)}{L} + \log_2 \frac{1}{\eta} + \mathcal{O}(1), \quad (8.7)$$

$$m_{\text{ctr}}(x, \eta) = -\frac{W_{-1}(-Lx)}{L} + \frac{1}{2} \log_2 \frac{1}{\eta} + \mathcal{O}(1). \quad (8.8)$$

The $\mathcal{O}(1)$ constants may be chosen uniformly for x in any sufficiently small interval $(0, x_0]$. **Result derived in this report.**

Proof. Equation (8.5) is the beginning of (5.5):

$$G(-u) = \exp\left(-\frac{u}{2} + \frac{u^2}{72} + \mathcal{O}(u^4)\right), \quad C(-u) = \exp\left(\frac{u^2}{72} + \mathcal{O}(u^4)\right).$$

Hence there are positive constants c_i, C_i and u_0 such that $c_1 u \leq \delta_{\text{raw}}(u) \leq C_1 u$ and $c_2 u^2 \leq \delta_{\text{ctr}}(u) \leq C_2 u^2$ for $0 < u < u_0$. Combine these inequalities with (8.2). Solving $2^{-h} \asymp \eta$ and $4^{-h} \asymp \eta$, then adding the base depth $\lambda(x)$ and absorbing the ceiling into $\mathcal{O}(1)$, gives (8.7)–(8.8). $\square$

Remark 8.2 (Why centering halves the precision surcharge). At the leading saddle scale, reaching relative tolerance η in the raw product requires roughly $\log_2(1/\eta)$ factors beyond λ . Removing the exactly known linear factor $e^{-u/2}$ makes the residual even, so only half as many additional factors are required. This is a concrete algorithmic consequence of the probability centering hidden in [theorem 5.1](#).

8.2 A finite-block endpoint evaluation protocol

Algorithm 8.3 (Lambert-aware finite-block product evaluation). Given small $x > 0$, target logarithmic correction order N , and tolerance η :

1. Compute $\lambda = -W_{-1}(-x \log 2) / \log 2$ and $s_0 = 2^\lambda = \lambda/x$.
2. Choose

$$m = \left\lceil \lambda + \frac{1}{2} \log_2 \frac{1}{\eta} + c_N \right\rceil,$$

where c_N is a small safety margin determined from [\(5.7\)](#).

3. Set $u = s_0/2^m$ and evaluate the finite digital sum

$$P_m(e^{-u}) = \sum_{n=0}^{2^m-1} \varepsilon_n e^{-nu}$$

or its product form.

4. Reconstruct $G(-s_0)$ from [\(5.8\)](#), using the first N Bernoulli corrections and the explicit remainder bound.

Warning 8.4. This protocol certifies the product evaluation at the leading Lambert scale. Computing $F(x)$ itself still requires the repository’s saddle inversion, including the nonconstant periodic phase and its derivatives. The result here supplies a new certified inner engine for that larger endpoint calculation; it does not suppress the periodic phase.

9 A Thue–Morse Barnes-type tower from Dirichlet trivial zeros

9.1 Input from the Atlas

The Thue–Morse Atlas establishes that, for $a > 0$, the shifted Dirichlet series

$$D(s, a) = \sum_{n \geq 0} \frac{\varepsilon_n}{(n+a)^s} \tag{9.1}$$

has an entire continuation $\tilde{D}(s, a)$, satisfies

$$\tilde{D}(s, a) = 2^{-s} \left[\tilde{D}\left(s, \frac{a}{2}\right) - \tilde{D}\left(s, \frac{a+1}{2}\right) \right], \tag{9.2}$$

and has trivial zeros

$$\tilde{D}(-r, a) = 0 \quad (r = 0, 1, 2, \dots). \tag{9.3}$$

It also supplies the Mellin representation

$$\tilde{D}(s, a) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} e^{-at} \mathcal{E}(t) dt, \quad \mathcal{E}(t) = \prod_{j \geq 0} (1 - e^{-2^j t}), \tag{9.4}$$

where the extraordinary flatness of $\mathcal{E}$ at zero makes the continuation entire [\[3, 9\]](#).

9.2 The derivative tower

Definition 9.1 (Thue–Morse gamma tower). For each integer $r \geq 0$ and $a > 0$, define

$$\Gamma_r^{\text{TM}}(a) = \exp\left(\partial_s \tilde{D}(-r, a)\right). \quad (9.5)$$

Theorem 9.2 (Mellin, dyadic, and differential laws). *The functions (9.5) satisfy:*

$$\log \Gamma_r^{\text{TM}}(a) = (-1)^r r! \int_0^\infty t^{-r-1} e^{-at} \mathcal{E}(t) dt, \quad (9.6)$$

$$\Gamma_r^{\text{TM}}(a) = \left(\frac{\Gamma_r^{\text{TM}}(a/2)}{\Gamma_r^{\text{TM}}((a+1)/2)} \right)^{2^r}, \quad (9.7)$$

$$\frac{d}{da} \log \Gamma_r^{\text{TM}}(a) = r \log \Gamma_{r-1}^{\text{TM}}(a) \quad (r \geq 1). \quad (9.8)$$

More generally, for $0 \leq k \leq r$,

$$\frac{d^k}{da^k} \log \Gamma_r^{\text{TM}}(a) = r^k \log \Gamma_{r-k}^{\text{TM}}(a). \quad (9.9)$$

Result derived in this report.

Proof. Near $s = -r$, the reciprocal gamma function has the expansion

$$\frac{1}{\Gamma(s)} = (-1)^r r! (s+r) + \mathcal{O}((s+r)^2).$$

The Mellin integral in (9.4) is finite there, so differentiation gives (9.6).

Differentiate (9.2) with respect to s and set $s = -r$. The derivative of 2^{-s} is multiplied by the bracket of values at $s = -r$, which vanishes by (9.3). Hence

$$\partial_s \tilde{D}(-r, a) = 2^r \left[\partial_s \tilde{D}\left(-r, \frac{a}{2}\right) - \partial_s \tilde{D}\left(-r, \frac{a+1}{2}\right) \right].$$

Exponentiation gives (9.7).

Finally, differentiation of the Dirichlet/Mellin continuation in the parameter gives

$$\partial_a \tilde{D}(s, a) = -s \tilde{D}(s+1, a).$$

Differentiate this identity in s :

$$\partial_a \partial_s \tilde{D}(s, a) = -\tilde{D}(s+1, a) - s \partial_s \tilde{D}(s+1, a).$$

At $s = -r$ with $r \geq 1$, the first term vanishes by (9.3), leaving $r \partial_s \tilde{D}(1-r, a)$. This proves (9.8); iteration gives (9.9). $\square$

Corollary 9.3 (The Woods–Robbins layer). *If $G_{\text{WR}}(a, b)$ denotes the convergent Thue–Morse product used in the Atlas, for which*

$$\log G_{\text{WR}}(a, b) = \partial_s \tilde{D}(0, b) - \partial_s \tilde{D}(0, a),$$

then

$$G_{\text{WR}}(a, b) = \frac{\Gamma_0^{\text{TM}}(b)}{\Gamma_0^{\text{TM}}(a)}. \quad (9.10)$$

Thus the known product is the ratio law at the base of an infinite dyadic differential tower.

Remark 9.4 (Why “Barnes-type”). The hierarchy is not asserted to be a classical Barnes multiple-gamma family. The name refers to the derivative ladder (9.8): higher levels integrate logarithms of lower levels, while the dyadic equation carries the automatic-sequence structure. This packaging appears to be absent from the inspected Atlas even though all analytic ingredients needed for its proof are present there.

10 Numerical verification and order diagnostics

All computations in this section used 100-decimal arithmetic. Product tails were extended far beyond the displayed precision. The calculations are reproducibility checks for the proved identities, not evidence on which the proofs depend.

10.1 Exact-identity residuals

Identity	level	argument	relative residual
oscillatory (3.1)	$m = 1$	$t = 0.37$	7.3×10^{-102}
oscillatory (3.1)	$m = 3$	$t = 2.125$	1.9×10^{-101}
oscillatory (3.1)	$m = 7$	$t = 5.75$	7.0×10^{-101}
oscillatory (3.1)	$m = 10$	$t = -3.4$	3.8×10^{-99}
Laplace (3.3)	$m = 4$	$s = 3.2$	5.6×10^{-101}
Laplace (3.3)	$m = 8$	$s = 11.5$	3.0×10^{-100}
Laplace (3.3)	$m = 12$	$s = 0.019$	9.4×10^{-96}

The residual level is consistent with accumulated arbitrary-precision rounding.

10.2 Observed convergence orders

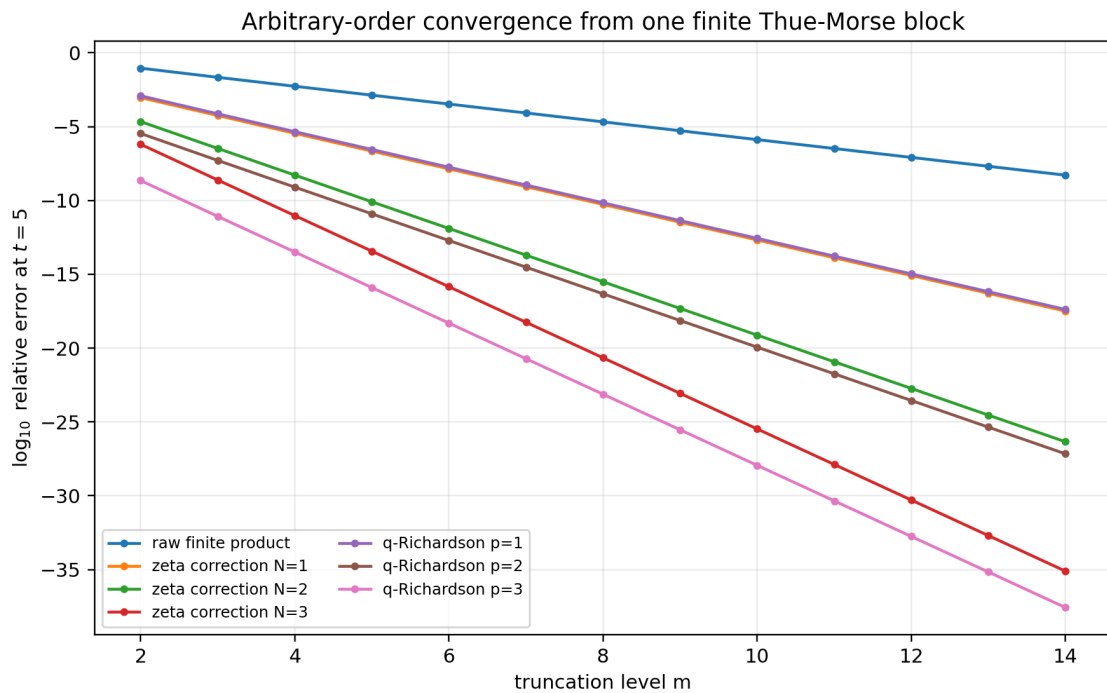


Figure 1: Relative errors at $t = 5$. Every zeta correction or Richardson degree raises the power of 4^{-m} by one, exactly as predicted by [theorems 4.5](#) and [6.2](#).

A least-squares fit to the final five points of each curve gives:

method	fitted order in 4^{-m}	proved order
raw Q_m	1.00000013	1
zeta correction $N = 1$	2.00000006	2
zeta correction $N = 2$	3.00000007	3
zeta correction $N = 3$	4.00000008	4
q -Richardson $p = 1$	2.00000014	2
q -Richardson $p = 2$	3.00000013	3
q -Richardson $p = 3$	4.00000013	4

10.3 Weak expansion check

For $\varphi(y) = e^y$, all derivatives equal e^y , so [theorem 7.3](#) becomes a direct moment correction to the finite moment-generating function. The relative errors are:

m	raw spline	through μ_2	through μ_4	through μ_6
2	3.465×10^{-3}	4.569×10^{-6}	3.311×10^{-9}	1.540×10^{-12}
4	2.170×10^{-4}	1.789×10^{-8}	8.105×10^{-13}	2.357×10^{-17}
6	1.356×10^{-5}	6.991×10^{-11}	1.979×10^{-16}	3.597×10^{-22}
8	8.477×10^{-7}	2.731×10^{-13}	4.832×10^{-20}	5.489×10^{-27}
10	5.298×10^{-8}	1.067×10^{-15}	1.180×10^{-23}	8.375×10^{-32}

Each added even moment raises the order by one power of 4^{-m} .

10.4 Endpoint-tail coefficients

The absolute errors in approximating $\log G(-u)$ by the exact linear term plus the first N even corrections are:

u	$N = 0$	$N = 1$	$N = 2$	$N = 3$
0.02	5.56×10^{-6}	3.70×10^{-12}	5.60×10^{-18}	1.04×10^{-23}
0.2	5.56×10^{-4}	3.70×10^{-8}	5.60×10^{-12}	1.04×10^{-15}
1	1.39×10^{-2}	2.31×10^{-5}	8.71×10^{-8}	4.03×10^{-10}
3	1.23×10^{-1}	1.81×10^{-3}	6.12×10^{-5}	2.54×10^{-6}

The deterioration as u approaches the exact radius 4π is governed by the certified bound [\(5.7\)](#).

10.5 Lambert-depth experiment

For $x = 10^{-8}$,

$$\lambda(x) = 31.555232088424067978\dots, \quad s_0(x) = 3.1555232088424068 \times 10^9.$$

At $h = 8$, for example, $m = 40$ and $u \approx 2.8699 \times 10^{-3}$; the raw tail error is 1.43×10^{-3} while the centered tail error is 1.14×10^{-7} .

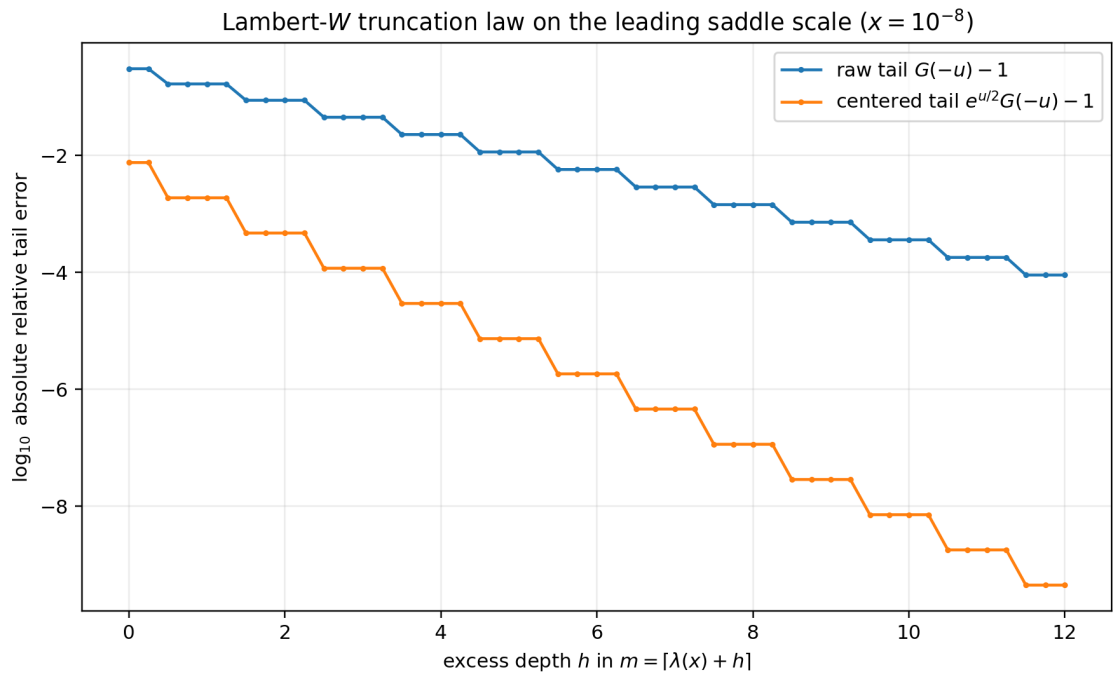


Figure 2: Tail error when $m = \lceil \lambda(x) + h \rceil$. Ceiling effects create the steps. The raw tail gains about one binary digit per extra factor; the centered tail gains about two, as proved in [theorem 8.1](#).

11 Consequences for the existing research program

11.1 A new exact-arithmetic/Fourier bridge

The Atlas treats the block polynomial P_m as a digital and algebraic object, while the Fourier-decay program treats dyadic sinc products through shell dynamics and lacunary sine products. Theorem 3.1 shows these are not merely related by analogy: P_m is the finite sinc product after one explicit normalization. This provides a direct translation dictionary:

finite digital statement	spectral statement
order- m zero of P_m at 1	removable t^{-m} normalization at the origin
Thue–Morse exponential sum	truncated up-function Fourier transform
Prouhet moment cancellation	high-order low-frequency cancellation
block length 2^m	product depth m
geometric error 4^{-m}	variance scale of omitted random tail

The shell-factorization results remain essential for large real t with m tied to the frequency shell. The present tail calculus addresses a different regime: once the first m factors have been isolated, it computes the remaining small-argument factor to arbitrary order. The two analyses are complementary rather than competing.

11.2 A new q -binomial role

The repository’s dyadic arithmetic naturally uses $q = 1/2$ because scales and evaluation points are dyadic. The present extrapolation naturally uses $q = 1/4$ because the centered error expansion is even. Gaussian binomials now appear in a second structural role:

$$\text{exact dyadic values: } q = \frac{1}{2}, \quad \text{spectral-tail acceleration: } q = \frac{1}{4}. \quad (11.1)$$

This distinction is conceptually useful. It prevents the q -binomial notation from being viewed as an incidental encoding of one exact formula; it reflects two different geometric lattices in the same theory.

11.3 A certified route from finite splines to up

The repeated-integration dossiers identify finite stages with compactly supported splines and prove convergence. The random-tail theorem strengthens the conceptual picture:

$$\text{limit law} = \text{finite spline law} + \text{scaled independent copy of the limit law}.$$

The weak expansion (7.7) then supplies every correction coefficient and a one-line remainder bound. For numerical integration against smooth observables, the result is immediately useful: one may integrate against the elementary finite spline u_m and add explicit derivative corrections, without ever sampling the limiting density directly.

11.4 A product engine for endpoint saddles

The endpoint theory’s difficult part is the periodically modulated saddle, not the mere existence of the product. Nevertheless, every saddle evaluation repeatedly calls the negative-Laplace product and its derivatives. Formula (5.8) suggests a hybrid implementation:

1. use the existing lower-Lambert phase and periodic saddle jets to choose s ;
2. choose m from the centered depth law (8.8);

3. evaluate the exact finite Thue–Morse block and a few Bernoulli corrections;
4. differentiate the finite factorization rather than the infinite product.

Because $u = s/2^m$ is small by construction, all differentiated correction series remain in a controlled analytic disk. This could turn the qualitative endpoint expansion into a certified high-precision algorithm with transparent complexity.

11.5 Formalization leverage

The main new identities have unusually short dependency chains. The finite bridges are finite-product algebra. Tail self-similarity is reindexing. The weak expansion is Taylor’s theorem plus independence and symmetry. The q -Richardson weights are finite Lagrange interpolation. Only the analytic logarithm and explicit zeta remainder require a substantial complex-analysis layer. This makes the package a plausible next formalization target.

12 Lean formalization roadmap

The finite-block calculus has been deliberately organized so that the algebraic core can be formalized before the analytic refinements. The following dependency table separates the short, high-confidence lemmas from the portions that require additions to the surrounding analysis library.

Target	Minimal ingredients	Suggested proof strategy and principal risk
Finite Thue–Morse product (2.1)	finite products, binary recursion, geometric sums	Prove $P_{m+1}(z) = P_m(z)(1 - z^{2^m})$ and identify coefficients by induction. This should be almost entirely algebraic.
Finite sinc bridge (3.1)	complex exponential identities, finite-product reindexing	Rewrite $1 - e^{2ix} = -2ie^{ix} \sin x$, then normalize powers of two. The only nuisance is a clean treatment of the removable value at $t = 0$.
Finite Laplace bridge (3.3)	real or complex exponentials, finite products	Substitute $z = e^{-s/2^m}$ in the block product and reindex. This is the simplest new theorem in the package.
Exact tail self-similarity (3.4)–(3.5)	convergence of infinite products, shifted indices	Establish a reusable infinite-product reindexing lemma. Existing repository convergence results should discharge the side conditions.
Euler–zeta logarithm (4.1)	local analytic logarithm, Euler’s sine product, normal convergence	This is the main library-level analytic obligation. One may alternatively prove the power series by differentiating the cotangent expansion and fixing the constant at zero.
Certified zeta remainder (4.6)	positivity, monotonicity of $\zeta(2r)$, geometric series	Once (4.3) exists, the bound is elementary and avoids delicate asymptotic machinery.
q -Richardson weights (6.4)–(6.7)	finite Lagrange interpolation, q -Pochhammer algebra	First formalize the interpolation characterization (6.5); derive the Gaussian-binomial form as a normalization lemma.
Random-tail law (7.2)	independent random variables, almost-sure summation, equality in law	Split and reindex the convergent random series. A characteristic-function proof may be shorter if the current probability library lacks convenient random-series infrastructure.
Weak expansion (7.7)	Taylor theorem with remainder, expectation inequalities, symmetry	Condition on the finite sum; use bounded support to make the remainder estimate immediate.
Lambert depth law (8.7)–(8.8)	branch W_{-1} , ceilings, local exponential estimates	The analytic estimate is elementary after the definition of λ ; the likely missing piece is a convenient formal API for the lower real Lambert branch.

Target	Minimal ingredients	Suggested proof strategy and principal risk
Thue–Morse gamma tower (9.6)–(9.9)	holomorphic parameter differentiation, reciprocal-gamma zero, Atlas Dirichlet continuation	Reuse the Atlas continuation theorem; prove the simple-zero coefficient of $1/\Gamma$ at each nonpositive integer; then differentiate the dyadic equation and the shift parameter identity.

A practical promotion sequence is therefore

$$\begin{array}{c} \text{finite bridges} \longrightarrow \text{tail reindexing} \longrightarrow \text{probabilistic splitting} \\ \longrightarrow q\text{-interpolation} \longrightarrow \text{analytic logarithms} \end{array}$$

The first four stages already yield exact identities and acceleration mechanisms. The final stage adds the sharp radius and explicit zeta remainder. This ordering minimizes the amount of new complex analysis that must be trusted at any one time.

Formalization obligation 12.1 (Canonical normalization theorem). The repository uses several Fourier-transform conventions across historical manuscripts. A promoted theorem should package the convention explicitly and derive all rescalings between F , up , Q_∞ , and the characteristic function of Y_∞ . Doing so would prevent sign and 2π factors from leaking into later shell or saddle arguments.

Formalization obligation 12.2 (Reusable geometric-tail API). The identity “tail equals a scaled copy of the whole object” occurs simultaneously for products, convolutions, random series, and cumulant generating functions. A small abstract library for geometrically self-similar tails would make the dyadic case a specialization rather than four separate proofs.

13 Frontier conjectures and next deductions

The proved results isolate several questions which are both sharper and more tractable than the broad open-ended prompts in the older frontier notebooks. The statements below are intentionally separated from the theorem package.

Conjecture 13.1 (Certified endpoint evaluator with near-optimal product depth). *There is a fully explicit algorithm which, for sufficiently small $x > 0$ and target relative error η , evaluates $F(x)$ using*

$$-\frac{W_{-1}(-x \log 2)}{\log 2} + \frac{1}{2} \log_2 \frac{1}{\eta} + \mathcal{O}(1) \quad (13.1)$$

product layers, together with a bounded-order periodic saddle jet, and whose final error is rigorously at most η . The coefficient $1/2$ in the precision surcharge is optimal among methods which first center the omitted product but do not cancel its quadratic cumulant.

The depth estimate itself is proved in [theorem 8.1](#). What remains is to propagate the product error, its differentiated versions, and the periodic saddle displacement through the complete inversion formula. The conjectured optimality should follow from the nonzero variance $\kappa_2 = 1/9$.

Conjecture 13.2 (Cumulant-cancelled endpoint hierarchy). *After removing the first N even cumulants of the centered tail in addition to the linear factor, the extra depth required for tolerance η is*

$$\frac{1}{2N+2} \log_2 \frac{1}{\eta} + \mathcal{O}_N(1). \quad (13.2)$$

Equivalently, the N -corrected finite block should require only one additional product layer for every $2N+2$ requested binary digits of tail accuracy.

This is already a theorem for the isolated product, by (5.7); the conjectural part is that the same saving survives uniformly through the full saddle inversion, including all needed derivatives and the periodic phase.

Conjecture 13.3 (Strong spline correction away from knots). *Fix $N \geq 0$ and a compact interval $K \subset (-1, 1)$. After excluding a neighborhood of the knots of the finite spline u_m whose width is comparable to 2^{-m} , the distributional expansion (7.10) admits a pointwise or Hölder-norm version through order N , with remainder $\mathcal{O}_K(4^{-(N+1)m})$ after the natural derivative scaling.*

A global pointwise statement cannot be obtained by naively differentiating u_m : finite splines have knots, whereas up is smooth. The random-tail convolution indicates two viable repairs. One may either retain the exact final mollifier $2^m \text{up}(2^m \cdot)$, or state the expansion on cells separated from their endpoints. Establishing the best norm and boundary layer is a concrete approximation-theoretic problem.

Conjecture 13.4 (Dual- q transmutation). *There is a direct coefficient-level transform taking a substantial class of the repository's $q = 1/2$ exact dyadic formulae into $q = 1/4$ spectral-tail acceleration identities, with the squaring $q \mapsto q^2$ explained by parity or centering rather than by an ad hoc substitution.*

Evidence is structural: the exact dyadic theory resolves single binary scales, while every centered Fourier or probability correction is even and therefore sees squared scales. A useful form of the conjecture would turn Gaussian-binomial coefficient extraction at $q = 1/2$ into geometric-grid interpolation at $q = 1/4$.

Conjecture 13.5 (Automatic Barnes hierarchy). *The functions Γ_r^{TM} possess canonical Weierstrass products and asymptotic normalizations for which the dyadic law (9.7) and differential ladder (9.8) uniquely characterize the hierarchy, analogously to the role of convexity or normalization in classical gamma-type characterizations.*

The Mellin formula gives positivity and complete-monotonicity information for suitable signed derivatives, while the dyadic equation supplies an automatic analogue of a multiplication formula. The remaining difficulty is to choose a normalization that removes the freedom to multiply by functions invisible to the differential recursion.

13.1 General radix and nonuniform tails

The base- b theorem (4.11) shows that the zeta–Lambert mechanism does not depend on binary arithmetic. What is special to base two is the simultaneous existence of the Thue–Morse block polynomial, the simple two-branch functional equation, and the Fabius/ up probability model. A next generalization is to replace the uniform summands by centered compactly supported variables X_k at scales b^{-k} . If their common characteristic function has a nonzero analytic germ at the origin, then

$$\log \frac{\Phi_\infty(t)}{\Phi_m(t)} = \sum_{r \geq 1} c_r t^r \frac{b^{-r(m+1)}}{1 - b^{-r}}.$$

Thus Lambert kernels are universal for geometric random tails; the zeta values in the present report are the special signature of the uniform law through Euler's sine product. The classification of digital substitutions whose finite block polynomial exactly realizes such a characteristic product is a natural interface between automatic sequences and compactly supported refinable laws.

14 Conclusion

The documentation corpus already supplied the large pieces: a careful separation of the Fabius and up -function normalizations, an exact dyadic arithmetic theory rich in q -binomial coefficients,

a probabilistic convolution model, a detailed analysis of the infinite sinc product, corrected lower-Lambert endpoint asymptotics, and a broad Thue–Morse formula atlas. The missing junction was finite-block calculus.

One polynomial now connects the principal domains:

$$P_m(z) = \sum_{n < 2^m} \varepsilon_n z^n$$

finite digital block
finite Fourier sinc product
finite negative-Laplace product

after explicit phase and scale normalizations. The omitted object is always a scaled copy of the full object. That observation produces exact all-order tails, zeta–Lambert coefficients, Gaussian-binomial extrapolation, random-tail deconvolution, and Lambert- W complexity laws from a common source rather than as isolated formulas.

The principal proved advances are:

1. exact oscillatory and decaying substitutions converting a length- 2^m Thue–Morse block into the first m factors of the Fourier and endpoint products;
2. a convergent all-orders logarithmic tail with exact radius and an explicit geometric remainder bound;
3. arbitrary-order corrected finite-block approximants and a separate q -Richardson scheme whose weights are Gaussian binomials at $q = 1/4$;
4. an exact independent-copy decomposition of the omitted random tail, yielding a certified weak differential expansion of the up-law in finite uniform splines;
5. raw, centered, and higher-correction product-depth laws on the canonical lower-Lambert saddle scale;
6. a Thue–Morse gamma tower obtained by differentiating the entire shifted Dirichlet function at its trivial zeros.

These results do not replace the repository’s sharp shell dynamics or periodically modulated endpoint saddle. They add a complementary local engine: after a finite block or finite shell has been isolated, the remainder is computable to arbitrary order with transparent constants. That engine is algebraically short, numerically stable after centering, and unusually suitable for formal verification.

A Coefficient and extrapolation tables

A.1 Zeta–Lambert and centered Laplace coefficients

Write

$$a_r = \frac{\zeta(2r)}{r\pi^{2r}(4^r - 1)}, \quad c_r = \frac{B_{2r}}{2r(2r)!(4^r - 1)}.$$

Then

$$\log \frac{Q_\infty(t)}{Q_m(t)} = - \sum_{r \geq 1} a_r t^{2r} 4^{-rm}, \quad \log C(-u) = \sum_{r \geq 1} c_r u^{2r}.$$

The first eight exact coefficients are:

r	a_r	c_r
1	1/18	1/72
2	1/2700	−1/43200
3	1/178605	1/11430720
4	1/9639000	−1/2467584000
5	1/478533825	1/490018636800
6	691/15688261338750	−691/64259118443520000
7	2/2092151286225	1/17138903336755200
8	3617/170727360353550000	−3617/11188788288130252800000

The numerators 691 and 3617 are the familiar Bernoulli numerators; here they enter through the uniform-tail cumulants rather than through an independent number-theoretic construction.

A.2 Analytic error germ

For

$$H_t(r) = \frac{1}{Q_\infty(t\sqrt{r})} = \sum_{n \geq 0} h_n(t) r^n,$$

the first coefficients are monomials in t^2 :

$$\begin{aligned} h_0(t) &= 1, \\ h_1(t) &= \frac{t^2}{18}, \\ h_2(t) &= \frac{31t^4}{16200}, \\ h_3(t) &= \frac{2347t^6}{42865200}, \\ h_4(t) &= \frac{1904369t^8}{1311675120000}, \\ h_5(t) &= \frac{3309760579t^{10}}{88561680752160000}. \end{aligned}$$

They are complete exponential Bell polynomials in $a_1 t^2, a_2 t^4, \dots$. Consequently, all leading constants in (6.10) are exact rational multiples of $Q_\infty(t) t^{2p+2}$.

A.3 Gaussian-binomial extrapolation weights

At $q = 1/4$, the first six rows are:

$$\begin{aligned}
 p = 0 : & \quad (1), \\
 p = 1 : & \quad \left(-\frac{1}{3}, \frac{4}{3}\right), \\
 p = 2 : & \quad \left(\frac{1}{45}, -\frac{4}{9}, \frac{64}{45}\right), \\
 p = 3 : & \quad \left(-\frac{1}{2835}, \frac{4}{135}, -\frac{64}{135}, \frac{4096}{2835}\right), \\
 p = 4 : & \quad \left(\frac{1}{722925}, -\frac{4}{8505}, \frac{64}{2025}, -\frac{4096}{8505}, \frac{1048576}{722925}\right), \\
 p = 5 : & \quad \left(-\frac{1}{739552275}, \frac{4}{2168775}, -\frac{64}{127575}, \frac{4096}{127575}, -\frac{1048576}{2168775}, \frac{1073741824}{739552275}\right).
 \end{aligned}$$

The alternating signs are unavoidable because the formula extrapolates from positive nodes $q^m, \dots, q^{m+p}$ to the endpoint 0. For numerical work at very high p , barycentric or compensated summation should be used rather than forming the rational coefficients in floating point.

B Reproducibility protocol

The numerical checks used direct high-precision arithmetic and did not fit coefficients that were subsequently inserted into proofs. The following pseudocode records the independent verification path.

Listing 1: Independent verification skeleton

```

set_precision(100)

def epsilon(n):
    return (-1) ** binary_digit_sum(n)

def P(m, z):
    return product(1 - z ** (2 ** j) for j in range(m))

def Q_finite(m, t):
    return product(sinc(t / (2 ** k)) for k in range(1, m + 1))

def Q_infinite(t, cutoff):
    return product(sinc(t / (2 ** k)) for k in range(1, cutoff + 1))

# Exact finite identities
for (m, t) in test_grid:
    lhs = P(m, exp(1j * t / (2 ** (m - 1))))
    rhs = ((-1j) ** m * t ** m * 2 ** (-m * (m - 1) / 2)
           * exp(1j * t * (1 - 2 ** (-m)))) * Q_finite(m, t)
    report(relative_residual(lhs, rhs))

# All-order tail and q-Richardson slopes
for m in depth_grid:
    exact = Q_infinite(t, large_cutoff)
    for N in correction_orders:
        approx = Q_finite(m, t) * exp(-sum(a(r) * t ** (2*r))

```

```

* 4 ** (-r*m)
for r in range(1, N + 1)))
save_error(m, N, relative_error(approx, exact))
for p in extrapolation_orders:
    approx = sum(weight(p, j) * Q_finite(m + j, t)
                  for j in range(p + 1))
    save_error(m, p, relative_error(approx, exact))
fit_final_log_slopes()

# Random-tail weak expansion, phi(y)=exp(y)
for m in depth_grid:
    finite_mgf = product(sinh(2 ** (-k)) / (2 ** (-k))
                        for k in range(1, m + 1))
    corrected = finite_mgf * sum(moment(2*j) * 4 ** (-j*m)
                                / factorial(2*j) for j in range(N + 1))
    compare_with_infinite_mgf(corrected)

```

The oscillatory and Laplace finite identities were tested at levels up to $m = 12$ with relative residuals between 10^{-96} and 10^{-102} . Regression was used only to diagnose the convergence exponent after the formulas had been proved; the fitted base-4 orders were 1, 2, 3, 4 to roughly seven significant digits for the raw, first-, second-, and third-order schemes. The figures embedded in this report were generated from the same data.

C Complete corpus ledger and deduplication map

C.1 Living documentation treated as authoritative

The audit covered every TeX manuscript in the living documentation tree. At the snapshot date, the principal paths and roles were:

Path or group	Role in the audit
Fabius_Function_and_Rvachev_Up /Fabius_Function_and_Rvachev_Up. tex	Primary synthesis: normalization, probability, Fourier theory, exact dyadic values, and corrected endpoint asymptotics.
FabiusFunction_Mathematical_Gl ossary/FabiusFunction_Mathemat ical_Glossary.tex	Terminology and notation cross-check.
Non_Elementarity_of_the_Fabius _Function/Non_Elementarity_of_ the_Fabius_Function.tex	Differential-algebraic and non-elementarity context.
fabius_lean_walkthrough/fabius _lean_walkthrough.tex	Map between human mathematics, Lean files, dependencies, and proof status.
non-formalized-research-frontie rs/non-formalized-research-fro ntiers.tex	Consolidated gap register and non-formalized frontier synthesis.
non-formalized-research-frontie rs/drafts/Thue_Morse_Formula_A tlas/Thue_Morse_Formula_Atlas. tex	Current comprehensive formula atlas; principal source for the block polynomial and shifted Dirichlet continuation.

Path or group	Role in the audit
non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/	Twelve active manuscripts: the initial wave, comparative and second-wave audits, a gentle guide, the asymptotic-product problem statement, successive waves 2–8, and the final resolution. They were compared rather than collapsed because their hypotheses and correction histories differ.
papers/AIF_76_21-38/	Original and corrected TeX transcriptions of Arias de Reyna’s compact-support paper.
papers/arXiv-1502.06738/	Source TeX for Aistleitner–Hofer–Larcher on parametric Thue–Morse sequences and lacunary trigonometric products.
papers/arXiv-1702.05442/	Source TeX for Arias de Reyna’s English translation of the 1982 compact-support article.
papers/arXiv-1702.06487/	Source TeX for Arias de Reyna’s arithmetic study of dyadic Fabius values.
papers/arXiv-2211.13570/	Source TeX for Tóth’s shifted and combined Thue–Morse Dirichlet series.
papers/ubiq15/ubiq15-corrected.tex	Corrected source for imported automatic-sequence background.

Within the Fourier-decay group, the inspected subdirectories were Rvachev_Up_Fourier_Decay-2, Rvachev_Up_Fourier_Decay-3, Comparative_Audit, Gentle_Guide, Second_Wave_Audit, asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions, rvachev_up_fourier_decay-1, and rvachev_up_fourier_decay-4 through rvachev_up_fourier_decay-8. Each contains one principal TeX manuscript.

C.2 Frozen archive represented through provenance

The archive README states that entries preserve the first-introduced historical document, that later edits are not separately archived, and that byte-identical copies already present in the living tree were pruned. Accordingly, the audit used the following unique historical entries as a provenance map rather than counting every renamed or duplicated copy as new mathematical evidence.

Archive category	Frozen entries
Early notes	FabiusAsymptotic; K-foldsummationoverthesignedThue–Morse sequence.
Standalone studies	Non_Elementarity_of_the_Fabius_Function; Small_Argument_Asymptotics; fabius-inverse-dyadic-closed-form.
Primary exposition	Fabius_Function_and_Rvachev_Up; Fabius_Function_and_Rvachev_Up-2; Fabius_Function_and_Rvachev_Up-3; Fabius_and_Rvachev_Up-2; Fabius_and_Rvachev_up; Fabius_function_and_up_function.
Primary supplements	Fabius_Function_Missing_Parts; Fabius_Function_Missing_Parts-2.
q-calculus	Fabius_q_Special_Functions; fabius-q-special-functions.
Lean walkthrough	Two frozen walkthrough variants recorded under the archive’s lean-walkthrough category.
Glossary	Two frozen glossary variants recorded under the archive’s glossary category.

Archive category	Frozen entries
Research frontiers	Fabius_Dyadic_Asymptotic_Bridge; Fabius_Dyadic_Formulae_and_Alternative_Representations; Fabius_Dyadic_Formulae_to_Asymptotics; Fabius_Dyadic_q_Connections; Fabius_Thue_Morse_Convergence_Rate; Fabius_Thue_Morse_Convergence_Rates; Primary_Exposition_Gap_Register; Repeated_Integration_and_Rvachev_Up; Rvachev_Up_from_Repeated_Integration; non-formalized-research-frontiers; Thue_Morse_Formula_Atlas-1; Thue_Morse_Formula_Atlas-2; Thue_Morse_Formula_Atlas-3.

This ledger explains the phrase “all TeX documents” operationally: every living manuscript was inspected, every active mutually auditing draft was retained, every imported TeX paper was included, and frozen historical variants were read through their explicit provenance and content map so that superseded copies did not distort novelty checks.

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